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PRE- BOARD EXAM-1 QUESTION PAPER

CLASS- X (2024-25)
SET-1

MATHEMATICS BASIC (241)

TIME: 3 hours **MAX.MARKS: 80**

General Instructions:

Read the following instructions carefully and follow them:
1. This question paper contains 38 questions.
2. This Question Paper is divided into 5 Sections A, B, C, D and E.
3. In Section A, Questions no. 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion- Reason based questions of 1 mark each.
4. In Section B, Questions no. 21-25 are very short answer (VSA) type questions, carrying 02 marks each.
5. In Section C, Questions no. 26-31 are short answer (SA) type questions, carrying 03 marks each.
6. In Section D, Questions no. 32-35 are long answer (LA) type questions, carrying 05 marks each.
7. In Section E, Questions no. 36-38 is case study-based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.
8. All Questions are compulsory. However, an internal choice in 2 Questions of section B, 2 Questions of section C and 2 Questions of section D has been provided. And internal choice has been provided in all the 2 marks questions of Section E.
9. Draw neat and clean figures wherever required.
10. Take π =22/7 wherever required if not stated.
11. Use of calculators is not allowed.

SECTION-A

This section consists of 20 questions of 1 mark each.

Q.1 The HCF of smallest prime number and the smallest composite number is

(a) 1 (b) 2 (c) 4 (d) 8

Q.2 If $P(x)$= ax^2+bx+c, then $\frac{-b}{a}$ is equal to

(a) 0 (b) 1 (c) Product of zeroes (d) Sum of zeroes

Q.3 The quadratic polynomial, the sum of whose zeroes is -4 and their product is 4, is

(a) x^2+4x+4 (b) x^2-4x+4 (c) x^2-4x-4 (d) $-x^2+4x+4$

Q.4 The Product of the zeroes of $-2x^2+kx+6$ is:

(a)1 (b) 2 (c) -3 (d) ¬4

Q.5 The pair of linear equations $2x + 3y = 5$ *and* $4x + 6y = 10$ *is*

(a) inconsistent (b) consistent (c) dependent and consistent (d) none of these

Q.6 If x =2 is a solution of the equation $x^2-5x+6k=0$, the value of k is

(a) 2 (b) 1 (c) 0 (d) 3

Q.7 In an AP if the common difference, d = -4 and the seventh term (a_7) is 4 then the first term is

(a) 32 (b) 28 (c) 24 (d) 20

Q.8 The distance of the point $(8, -6,)$ from origin is :

(a) 10 (b) -6 (c) 8 (d) 6

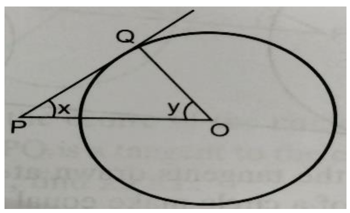
Q.9 The ratio in which the line $3x + y - 9 = 0$ divides the line segment joining the points (1,3) and (2,7) is

(a) 3:2 (b) 2:3 (c) 4:3 (d) 3:4

Q.10 Three vertices of a parallelogram ABCD are A(1,4), B(−2,3) $and\ C(5,8)$. The ordinate of the fourth vertex D is

(a) 8 (b) 9 (c) 7 (d) 6

Q.11 In the given figure, PQ is a tangent to the circle with center O. If $\angle OPQ = x, \angle POQ = y,$ then $x + y$ is:



(a) 45° (b) 90° (c) 60° (d) 180°

Q.12 If $3\cot\theta = 2, then\ the\ value\ of\ tan\theta$ is

(a) $\frac{2}{3}$ (b) $\frac{3}{\sqrt{13}}$ (c) $\frac{3}{2}$ (d) $\frac{2}{\sqrt{13}}$

Q.13 The value of $sin^2 60° + 2tan45° - cos^2 30°$ is

(a) 1 (b) 3 (c) 0 (d) None of these

Q.14 The diameter of the hemisphere whose curved surface area is 308 cm^2 is

(a) 7 cm (b) 0.7 cm (c) 14 cm (d) 70 cm

Q.15 A solid cylinder of radius r and height h is placed over other cylinder of same height and radius. The total surface area of the shape so formed is

(a) $4\pi rh + 4\pi r^2$ (b) $2\pi rh + 4\pi r^2$ (c) $4\pi rh + 2\pi r^2$ (d) $2\pi rh + 2\pi r^2$

Q.16 If the Mode and Mean of a distribution are 14 and 20respectively, then the Median of the distribution is:

(a) 18 (b) 17 (c) 16 (d) 15

Q.17 In the following distribution:

Height (in cm)	100-115	115-130	130-145	145-160	160-175
Number of Students	15	13	11	10	11

The sum of the upper limit of the median class and lower limit of the modal class is

(a) 230 (b) 260 (c) 245 (d) 275

Q.18 When a die is thrown, the probability of getting a prime number is

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) 0

Directions: In question number 19 and 20 a statement of Assertion (A) is followed by a statement of Reason(R) Choose the correct option.

(a) Both, Assertion(A) and Reason(R) are true and Reason(R) is correct explanation of Assertion (A)

(b) Both, Assertion(A) and Reason(R) are true and Reason(R) is not correct explanation for Assertion (A)

(c) Assertion(A) is true but Reason(R) is false.

(d) Assertion(A) is false but Reason(R) is true.

Q.19 Assertion(A): A card is drawn at random from a well shuffled pack of 52 playing cards. The probability of getting neither a red card nor a queen is $\frac{24}{52}$ i.e. $\frac{6}{13}$

Reason(R): A month is selected at random from a year. The probability that it is March or June is $\frac{1}{6}$

Q.20 Assertion(A): Two players Surya and Virat play a tennis match. The probability of Virat winning the match is 0.78 and that of Surya winning the match is 0.22.

Reason(R): The sum of probabilities of two complementary events is 1.

SECTION-B

This section consists of 5 questions of 2 marks each.

Q.21 If product of two numbers is 3691 and their LCM is 3691, then find their HCF.

OR

Check whether 7^n can end with the digit 0 for any natural number n.

Q.22 Given the linear equation $4x + 6y - 16 = 0$, write another linear equation in two variables such that the geometrical representation of the pair so formed is:

(i) Parallel lines

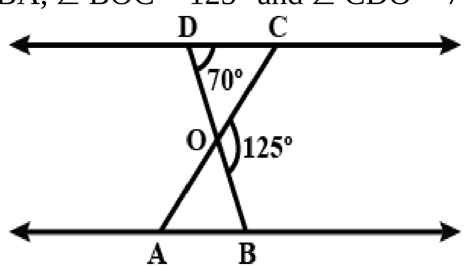
(ii) Intersecting lines

Q.23 E and F are points on the sides PQ and PR respectively of a Δ PQR. State whether EF || QR

PE = 5 cm, QE = 4.5 cm, PF = 10 cm and RF = 9 cm

OR

In given figure, Δ ODC ~ Δ OBA, $\angle$ BOC = 125° and $\angle$ CDO = 70°. Find $\angle$ OAB. And $\angle$ DOC



Q.24 If $\sin A = \frac{3}{4}$ calculate cos A and tan A.

Q.25 Rohit Sharma tosses two different coins simultaneously. Find the probability of getting at least one head.

SECTION-C

This section consists of 6 questions of 3 marks each.

Q.26 Prove that $\sqrt{2}$ is an irrational.

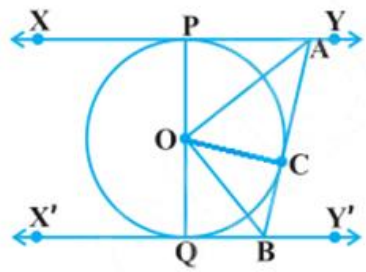
Q.27 Five years ago, Nuri was thrice as old as Sonu. Ten years later, Nuri will be twice as old as Sonu. How old are Nuri and Sonu?

Q.28 Find the coordinates of the point which divides the line segment joining the points (4, – 3) and (8, 5) in the ratio 3 : 1 internally.

OR

Find a relation between x and y such that the point (x, y) is equidistant from the points (7, 1) and (3, 5).

Q.29 In figure given below, XY and X′Y′ are two parallel tangents to a circle with center O and another tangent AB with point of contact C intersecting XY at A and X′Y′ at B. Prove that ∠ AOB = 90°.



Q.30 Prove that $\frac{tanA}{1-cotA} + \frac{cotA}{1-tanA}$ = sec A cosec A +1

OR

Prove that $(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 = 7 + \tan^2\theta + \cot^2\theta$

Q.31 A Gulab Jamun, contains sugar syrup up to about 30% of its volume. Find approximately how much syrup would be found in 45 Gulab jamuns, each shaped like a cylinder with two hemispherical ends with length 5 cm and diameter 2.8 cm.

SECTION-D

This section consists of 4 questions of 5 marks each.

Q.32 In a circle of radius 21 cm, an arc subtends an angle of 60° at the center. Find:

(i) Area of the sector formed by the arc

(ii) Area of the segment formed by the corresponding chord.

Q.33 Two poles of equal heights are standing opposite each other on either side of the road, which is 80 m wide. From a point between them on the road, the angles of elevation of the top of the poles are 60° and 30°, respectively. Find the height of the poles and the distances of the point from the poles.

Q.34 Find the 31st term of an AP whose 11th term is 38 and the 16th term is 73.

OR

Find the sum of first 51 terms of an AP whose second and third terms are 14 and 18 respectively.

Q.35 State and prove Basic Proportionality theorem (BPT).

OR

If AD and PM are medians of triangles ABC and PQR, respectively where Δ ABC ~ Δ PQR, prove that $\frac{AB}{PQ} = \frac{AD}{PM}$

SECTION-E

This section consists of 3 Case –Study Based Questions of 4 marks each.

Q.36 During the annual sports meet in a school; all the athletes were very enthusiastic. They all wanted to be the winner so that their house could stand first. The instructor noted down the time taken by a group of students to complete a certain race. The data recorded is given below:

Time (In seconds)	Number of Students
0-20	1
20-40	4
40-60	3
60-80	7
80-100	5

Based on the above information, answer the following questions:

(i) Find the median class of the given data. (1 mark)

(ii) What is the class mark of the modal class? (1 mark)

(iii) Find the mode of the given data. (2 marks)

OR

Find the median time (in seconds) of the given data.

Q.37 There is a wide range of concepts about boats and streams problems. Some of the major concepts related to boats and streams are mentioned below:

(1) Stream – A stream is used to refer to running river water.

(2) Upstream – When the boat rows or moves in the opposite direction of the flow of the stream, it is said to be moving upstream. The boat's net speed, in this case, is upstream speed.

(3) Downstream – When the boat rows or moves along the direction of the flow of the stream, it is said to be moving downstream. The boat's net speed, in this case, is downstream speed.

(4) Still Water – Still water is the assumption of water being still or stationary. This happens when the speed of the water is zero.

The speed of a motor boat is 20 km/hr. For covering the distance of 15 km the boat took 1 hour more for upstream than downstream.

Based on the above information, answer the following questions:

(i) If speed of the stream be x km/hr. then what will be the speed of the motorboat in upstream? (1 mark)

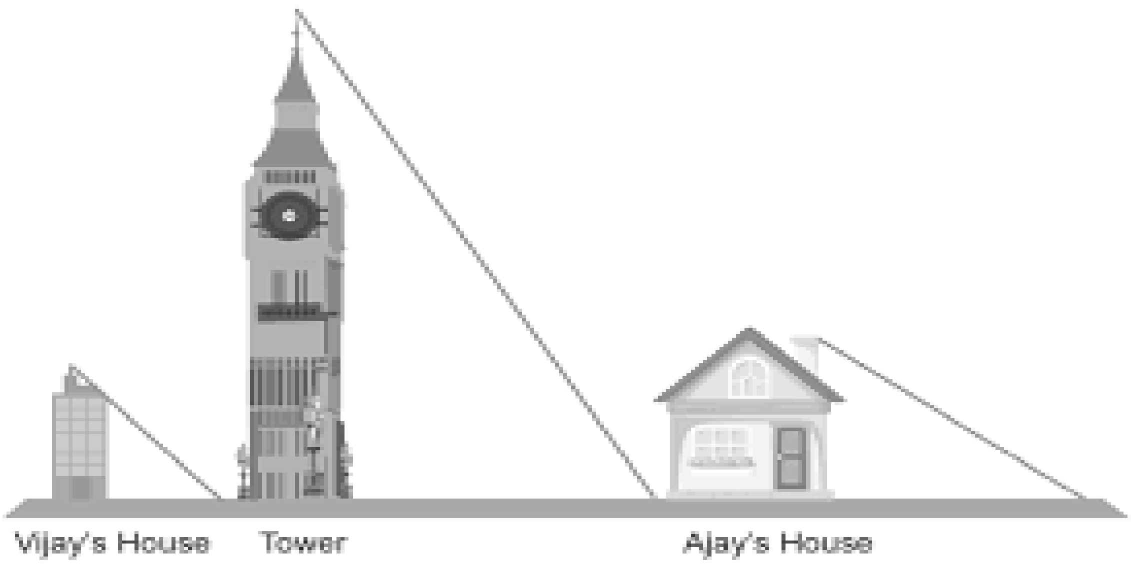
(ii) What is the relation between speed, distance, and time?? (1 mark)

(iii) What is the correct quadratic equation for the speed of the current? (2 marks)

OR

What is the speed of current?

Q.38 Vijay is trying to find the average height of a tower near his house. He is using the properties of similar triangles. The height of Vijay's house if 20 m when Vijay's house casts a shadow 10 m long on the ground. At the same time, the tower casts a shadow 50 m long on the ground and the house of Ajay casts 20 m shadow on the ground.



Based on the above information, answer the following questions:

(i) What is the height of the tower? (1 mark)
(ii) What will be the length of the shadow of the tower when Vijay's house casts a shadow of 12 m? (1mark)
(iii) When the tower casts a shadow of 40 m, same time what will be the length of the shadow of Ajay's house?

OR

When the tower casts a shadow of 40 m, same time what will be the length of the shadow of Vijay's house? (2 marks)